

Darwin Liao

Software Developer | M.Sc. Computer Science

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PROFESSIONAL SUMMARY

Software developer and recent M.Sc. Computer Science graduate with experience delivering React and Spring Boot applications. Background in REST APIs, access control, automated testing, AWS deployment, cybersecurity, and applied machine learning.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, SQL, C, HTML, CSS

Frameworks: React, Spring Boot, Next.js, Jest, Spring Test, .NET, Qt, TensorFlow, scikit-learn

Cloud, Data, and Systems: AWS, Azure, Terraform, CI/CD, Git/GitHub, PostgreSQL, REST APIs, SCADA, IEC 61850, Modbus, TCP/IP

PROFESSIONAL EXPERIENCE

Fullstack Developer | Intellijoint Surgical, Kitchener, ON

Apr 2023 - Aug 2024

- Delivered 100+ development tickets for a surgical pre-planning application built with React and Spring Boot, collaborating with 3 developers and 2 QA testers.
- Built reusable React components and workflows, including text inputs, context menus, read-only states, and keyboard shortcuts, and implemented case sharing with 4 permission levels.
- Developed concurrent-access functionality supporting up to 10 users per surgical case.
- Created a reusable Spring framework for rate limiting and request-size enforcement, protecting 40+ REST endpoints.
- Added Jest and Spring Test coverage for every delivered ticket, React component, and API endpoint.

Software Engineer | General Electric Grid Solutions, Markham, ON

Apr 2019 - Aug 2019

- Expanded an automated firmware-testing suite for substation protection relays, validating communication and security functionality across multiple relay models.
- Enhanced a universal testing and report-generation application used for 370+ recurring firmware-validation procedures, reducing manual effort and resource use.
- Tested relay communications in SCADA environments using IEC 61850, Modbus, and TCP/IP.
- Ported open-source C#, C++, and Java libraries to .NET, improving compatibility with the existing codebase and supported relay models.

EDUCATION

M.Sc. Computer Science, Distributed Systems | Western University, London, ON

Sep 2024 - Aug 2026

B.Sc. Computer Science | Western University, London, ON

Sep 2019 - Apr 2024

SELECTED RESEARCH AND PROJECTS

OTA-Sentinel | Master's Research | Python, TensorFlow, scikit-learn, AAOS

- Built a lightweight static anomaly-screening pipeline for automotive OTA files using byte-level features and PCA, Isolation Forest, VAE, f-AnoGAN, and Reptile-based adaptation.
- Evaluated vendor IVI and AAOS-derived update files against ELF/APK malware and controlled modifications, reaching up to 0.99 AUROC and measuring cross-domain transfer and runtime cost.

Project Lumi | Fantasy League and Event Hub | TypeScript, Python, PostgreSQL, AWS

- Built and deployed an Eternal Return fantasy esports platform for approximately 1,000 users with Discord authentication, PostgreSQL persistence, external game API integration, and AWS EC2 hosting.
- Created leaderboard, community bounty, and video-submission workflows for recurring esports events.

PUBLICATION

Domain-Adaptive Static Anomaly Screening for In-Vehicle Infotainment OTA Updates

36th IEEE Int. Conference on Collaborative Advances in Software and Computing (CASCON), 2026 | To Appear

Lightweight Static File Analysis for Pre-Installation Anomaly Screening in Automotive OTA Updates

Master's Thesis, 2026 | <https://uwo.scholaris.ca/items/4ed13b15-40a4-4b97-8a9e-3e3bf9f5b9b3>

OTArmor: Securing Automotive Over-the-Air Updates Against Malware Using Generative Modeling

9th Cyber Security in Networking Conference (CSNet), 2025 | doi.org/10.1109/CSNet67572.2025.11288195